

Dear Editor:

I am pleased to submit the manuscript titled "Superellipsoid-Constrained Bidirectional RRT: Path Planning with Adaptive Asymmetric Dual-Super-Ellipsoidal Constrained Sampling" for consideration for publication in *Industrial Robot: The International Journal of Robotics Research and Application*.

This research article presents SC-RRT, a novel sampling-based path planning algorithm that addresses key limitations of existing methods, including excessive invalid samples, slow convergence, and reduced success rates in high-obstacle-density environments. The proposed algorithm integrates two tightly coupled core mechanisms: an Adaptive Asymmetric Dual-Ellipsoid Constrained Sampling (ADCS) strategy and a Search-State Feedback Online Regulation (SSFOR) mechanism. Within a bidirectional parallel search framework guided by a dynamically updated intermediate meeting point, ADCS constructs independently scalable asymmetric hyper-ellipsoidal informed subsets on both the start and goal sides, concentrating sampling toward feasible regions with greater improvement potential. The SSFOR mechanism implements closed-loop regulation of the hyper-ellipsoid expansion factor and heuristic sampling probability based on the historical evolution of the optimal path cost, enabling the informed subset to evolve continuously with the search progress while suppressing redundant expansion.

Comparative simulation experiments conducted in both two-dimensional and three-dimensional environments with varying obstacle densities demonstrate that SC-RRT achieves a path efficiency of 0.9553, reduces path length by approximately 10.5% compared with Dynamic-RRT, and requires only 16.2% of the planning time of Informed RRT\*, with node count reduced by 94.2%. These results confirm that SC-RRT achieves an optimal overall balance among path quality, computational efficiency, and planning stability, without requiring prior parameter tuning. The modular ADCS and SSFOR components can furthermore be independently integrated into other RRT-based frameworks.

We believe this work is highly relevant to your readership as it directly contributes to the precise motion planning and control of industrial robots in cluttered environments. The algorithm's stability and lack of requirement for prior parameter tuning make it particularly suitable for real-world engineering applications.

I confirm that this manuscript is an original work, has not been published elsewhere, and is not currently under consideration by another journal. All authors have reviewed and approved the manuscript for submission.

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Thank you for considering our manuscript. We look forward to your feedback.

Sincerely,

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